

Physics-Informed Machine Learning for PMSM Stator Winding Temperature Prediction

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Abstract—Permanent Magnet Synchronous Motors (PMSMs) are widely deployed in electric vehicles, industrial automation, and robotics because of their efficiency and high power density. However, their safe operation critically depends on accurate stator winding temperature estimation, since thermal overload can degrade insulation, reduce efficiency, and shorten motor lifetime. Purely data-driven models often perform well on familiar operating conditions but can fail to generalize under unseen profiles or high-stress regimes.

This report presents a Physics-Informed Machine Learning (PIML) framework for PMSM stator winding temperature prediction. The study compares four models: Random Forest, a conventional Deep Neural Network (DNN), a Physics-Guided DNN, and a Residual PIML model that learns the error remaining after a physics-based thermal estimate. The framework additionally includes physics-guided feature engineering, profile-based validation, stress testing under high current and high mechanical power, and uncertainty-aware analysis. On the evaluated dataset, the proposed Residual PIML achieved the best overall accuracy with MAE = 1.1719°C, RMSE = 1.6336°C, and $R^2 = 0.9972$, outperforming both conventional ML and standard DNN baselines. The Physics-Guided DNN also improved markedly over the DNN baseline, confirming that incorporating thermal physics into learning is beneficial for accurate and robust motor temperature prediction.

Index Terms—Physics-Informed Machine Learning, PMSM, temperature prediction, predictive maintenance, residual learning, deep learning, Random Forest, thermal modeling.

I. INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) are a core component of modern electro-mechanical systems. They are preferred in many high-performance applications because they deliver high torque density, strong efficiency, and precise controllability. Despite these advantages, PMSM operation is strongly constrained by thermal behavior. The stator winding temperature is particularly important because it directly affects insulation aging, loss mechanisms, and reliability.

The practical problem addressed in this work is simple but high value: given electrical, mechanical, and thermal sensor readings from a PMSM, estimate the stator winding temperature as accurately as possible. In a production setting, this estimate can support condition monitoring, thermal protection, fault prevention, and predictive maintenance.

The reason this problem is non-trivial is that temperature dynamics are not purely statistical. They are governed by the interaction of electrical losses, mechanical loading, heat

transfer, and thermal inertia. A model that ignores this structure may fit the training data but still behave poorly under shifted operating conditions. That is exactly why this work uses Physics-Informed Machine Learning (PIML): the model is not asked to infer everything from data alone; it is guided by known physical behavior.

A. Research Questions

This report is organized around the following research questions:

- 1) Can a classical machine learning model predict PMSM winding temperature accurately?
- 2) Does a deep neural network improve over classical machine learning?
- 3) Does injecting physics into the learning process improve accuracy and robustness?
- 4) Does residual physics-based modeling outperform direct temperature regression?
- 5) Can the proposed approach remain stable under unseen profiles and stress conditions?

B. Project Repository

The implementation associated with this report is available in the [project GitHub repository](#).

C. Why This Study Matters

The answer to the above questions determines whether a temperature predictor is useful only on a benchmark dataset or also credible in real-world deployment. For a motor health monitoring system, the second is the only one that matters. The work therefore does not stop at a single test split. It evaluates the models under unseen profiles, high current, and high mechanical power conditions, which is necessary to demonstrate practical value rather than just benchmark performance.

II. DATASET AND PROBLEM SETUP

The PMSM dataset contains electrical and thermal measurements collected under multiple operating profiles. The dataset used in this work is the [Kaggle Electric Motor Temperature dataset](#). The input variables used in this report include direct-axis current i_d , quadrature-axis current i_q , direct-axis voltage u_d , quadrature-axis voltage u_q , motor speed, torque, coolant

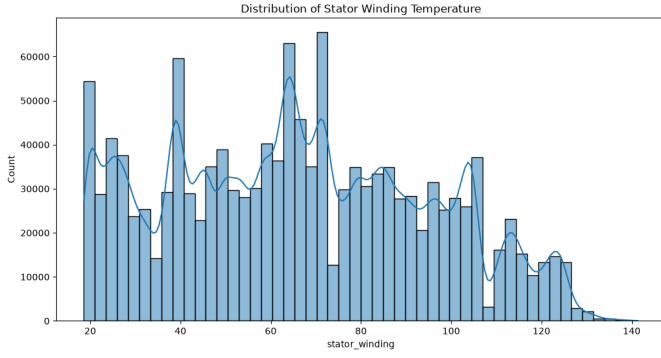


Fig. 1. Distribution of the stator winding temperature target. The spread of values indicates that the task is a continuous thermal-regression problem across both moderate and high-temperature regimes; any model that performs well must therefore capture the full operating envelope instead of optimizing only around the most frequent temperature range.

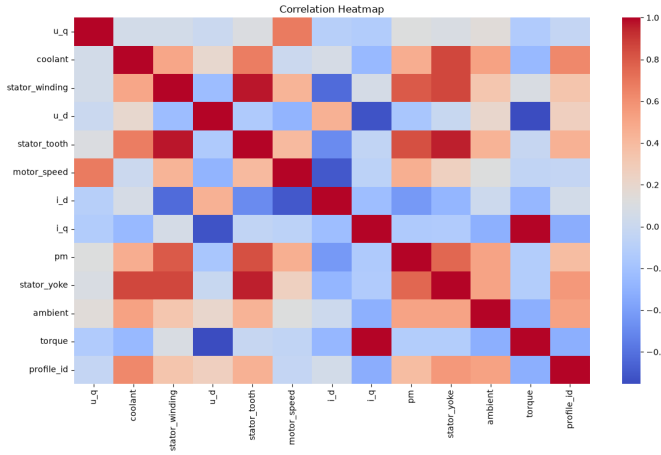


Fig. 2. Correlation analysis between raw variables, engineered physical features, and stator winding temperature. Strong associations involving current magnitude, torque, speed, and coolant temperature support the central modeling assumption that thermal behavior is driven by coupled electrical loading and cooling-boundary conditions rather than by isolated sensor channels.

temperature, and profile identifiers. The target variable is stator winding temperature.

A. Why These Variables Were Used

Each input carries physical meaning. Current is related to copper losses, voltage contributes to electrical power behavior, torque and speed represent loading conditions, and coolant temperature captures the cooling boundary condition. The profile identifier is not used as a direct predictive variable in the main training pipeline; instead, it is used for evaluation under profile-wise generalization tests so that model robustness can be verified on unseen operating regimes.

B. Feature Engineering Philosophy

The goal of feature engineering here is not to inflate dimensionality; it is to expose physically meaningful signals that are hidden in the raw variables. A temperature model should know not only the current components themselves, but also

the magnitude of current, the approximate electrical power, and the mechanical power demand. These derived features are physically interpretable and help the model learn relationships that are otherwise difficult to discover from raw inputs alone.

III. PHYSICS BACKGROUND

The choice of a physics-informed model is motivated by thermal energy balance in electric machines. A simplified but useful description is that copper losses generate heat, and the resulting temperature is governed by the balance between heat generation and heat dissipation.

$$P_{cu} = I^2 R \quad (1)$$

where P_{cu} is the copper loss, I is the phase current, and R is the winding resistance. In practice, the exact resistance may vary with temperature, but the fundamental quadratic dependence on current remains valid and is the reason current magnitude is such an important predictor.

A generic lumped thermal model can be written as

$$C \frac{dT}{dt} = P_{loss} - h(T - T_{amb}), \quad (2)$$

where C is thermal capacitance, h is a heat transfer coefficient, T is winding temperature, and T_{amb} is ambient or cooling temperature. This equation explains why motor temperature exhibits thermal inertia: it does not jump instantly, but evolves smoothly over time.

A. Why Physics Helps

A data-only model must infer these relationships from examples. A physics-informed model starts with the fact that current increases heating, cooling temperature influences thermal equilibrium, and temperature changes should be smooth. This makes the learning problem better conditioned and improves generalization, especially under data shifts.

IV. METHODOLOGY

This section explains what was implemented, why it was implemented, how it was used, and what each component contributed to the final results.

A. Data Preprocessing

The first implementation step was cleaning the raw data and preparing it for training. This included handling missing values if present, separating inputs from the target, and standardizing numerical features for the neural network models.

a) *Why standardization was used.*: Neural networks train more reliably when inputs are on comparable scales. Without scaling, a feature with a large numeric range can dominate the gradient updates and slow or destabilize optimization. Random Forest does not require scaling, but DNN-based models do.

b) *How it was used.*: A scaler was fit on the training data only and then applied to validation and test data. This avoids information leakage. The same scaling pipeline was reused for the Physics-Guided DNN and Residual PIML models.

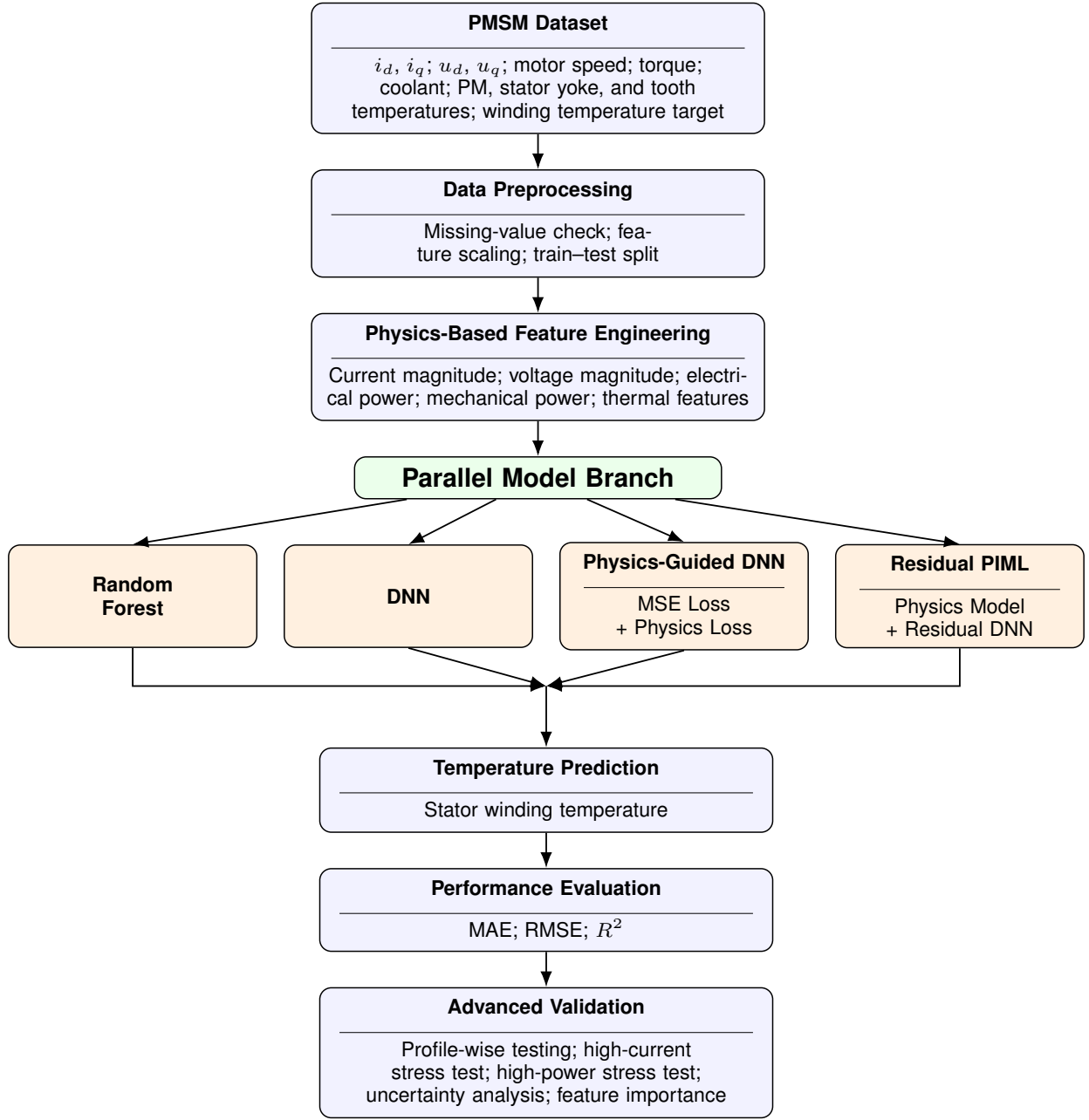


Fig. 3. Detailed architecture of the PMSM stator winding temperature prediction framework. The diagram shows how raw electrical, mechanical, and thermal measurements are preprocessed, transformed into physics-informed features, evaluated through four model branches, and validated through standard metrics, profile-wise testing, stress tests, uncertainty analysis, and feature-importance evidence.

B. Physics-Guided Feature Engineering

Several physically meaningful features were derived from the raw signals.

$$I_{mag} = \sqrt{i_d^2 + i_q^2} \quad (3)$$

a) *Why it was used.*: Current magnitude is more informative than individual current axes for heating because copper loss is driven by current amplitude.

b) *How it was used.*: This feature was included in the analysis and was also useful for interpreting high-current stress behavior.

$$V_{mag} = \sqrt{u_d^2 + u_q^2} \quad (4)$$

c) *Why it was used.*: Voltage magnitude is a compact representation of the electrical operating point.

d) *How it was used.*: It was used as an auxiliary derived feature to reflect electrical activity more directly than the axis-wise voltages alone.

$$P_{elec} = I_{mag} V_{mag} \quad (5)$$

e) *Why it was used.*: Electrical power is closely related to heat generation.

f) *How it was used.*: This feature improves the physical interpretability of the input space.

$$P_{mech} = \tau \omega \quad (6)$$

where τ is torque and ω is motor speed.

g) *Why it was used.*: Mechanical power captures the load condition of the motor, which affects thermal loading.

h) *How it was used.*: This feature becomes especially meaningful in the stress tests, where the model is evaluated under high mechanical power.

C. Baseline Model 1: Random Forest

Random Forest was used as a strong classical machine learning baseline.

a) *Why it was selected.*: For tabular regression problems with mixed nonlinear interactions, Random Forest is a reliable benchmark. It is robust, easy to train, and can capture nonlinearities without heavy tuning.

b) *How it was used.*: The model was trained on the processed input features using a standard ensemble-regression setup. Because tree-based methods do not require feature scaling, the baseline is relatively simple but powerful.

c) *What it showed.*: The Random Forest performed well with MAE = 1.6032°C and $R^2 = 0.9942$, demonstrating that the dataset contains learnable nonlinear structure even without physics constraints.

D. Baseline Model 2: Conventional DNN

A fully connected deep neural network was trained directly on the input features.

a) *Why it was selected.*: A DNN is a flexible function approximator that can model complex interactions if the data and training process are suitable.

b) *How it was used.*: The network was trained using standard regression loss on scaled inputs.

c) *What it showed.*: The DNN baseline performed poorly relative to Random Forest in this implementation, with MAE = 4.7391°C and $R^2 = 0.9668$. This result is important because it shows that a purely data-driven neural network is not automatically better than a classical ensemble for structured thermal data.

E. Model 3: Physics-Guided DNN

The Physics-Guided DNN is the first physics-aware model in the pipeline.

a) *Why it was selected.*: The purpose is to inject domain knowledge into the optimization process so that the network is not free to learn physically unreasonable mappings.

b) *How it was used.*: The model was trained with a combined loss function:

$$L = L_{MSE} + \lambda L_{physics}, \quad (7)$$

where L_{MSE} penalizes prediction error and $L_{physics}$ penalizes physically inconsistent behavior.

c) *What it showed.*: The Physics-Guided DNN achieved MAE = 1.2468°C, RMSE = 1.6021°C, and $R^2 = 0.9973$. This is the first strong proof that physics improves prediction quality.

F. Model 4: Residual PIML

Residual PIML is the main novelty of this work.

a) *Why it was selected.*: Rather than forcing a neural network to learn the entire temperature mapping from scratch, residual learning decomposes the problem into a physics estimate plus a learned correction term.

b) *How it was used.*: The final prediction is written as

$$T_{pred} = T_{physics} + T_{residual}. \quad (8)$$

The term $T_{physics}$ gives a coarse thermal estimate based on known domain relationships, while the neural network learns only the remaining error.

c) *Why this is a better formulation.*: This reduces the learning burden on the network, improves interpretability, and typically improves generalization because the network learns a residual rather than the full state.

d) *What it showed.*: Residual PIML achieved the best MAE of 1.1719°C and the best overall balance of accuracy and robustness. This is the strongest evidence that physics-guided residual learning is the most effective configuration in the project.

V. EVALUATION PROTOCOL

The evaluation was not limited to a random test split. A strong report needs to answer whether the model is genuinely useful under new conditions.

A. Standard Hold-Out Evaluation

The first test was a conventional train-validation-test split. This allows direct comparison between models using the same data partition.

a) *Why it was used.*: It is the most basic and widely understood evaluation protocol.

b) *How it was used.*: All models were tested on the same unseen test set, and performance was measured using MAE, RMSE, and R^2 .

B. Profile-Wise Generalization

The dataset contains multiple operating profiles. A profile-wise test was used to check whether the models generalize to unseen motor regimes.

a) *Why it was used.*: Random splits can overestimate performance because samples from the same operating regime may appear in train and test sets. Profile-wise evaluation is stricter and more realistic.

TABLE I
OVERALL MODEL PERFORMANCE

Model	MAE (°C)	RMSE (°C)	R^2
Random Forest	1.6032	2.3503	0.9942
DNN	4.7391	5.6513	0.9668
Physics-Guided DNN	1.2468	1.6021	0.9973
Residual PIML	1.1719	1.6336	0.9972

b) How it was used.: The model was evaluated on selected profiles separately, and profile-wise MAE was computed.

C. Stress Testing

Two stress regimes were studied: high current and high mechanical power.

a) Why it was used.: Thermal failures are most likely under high load. A useful temperature predictor must remain stable in these regimes.

b) How it was used.: Subsets of the test data corresponding to high current and high mechanical power were isolated and evaluated independently.

D. Metrics

The following metrics were used throughout the report.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (9)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (10)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (11)$$

MAE measures average absolute deviation, RMSE penalizes large errors more strongly, and R^2 measures how much variance is explained by the model.

VI. MAIN RESULTS

A. Overall Comparison

a) Interpretation.: Random Forest is a strong benchmark, but the Physics-Guided DNN improves over it by learning with physical constraints. Residual PIML achieves the lowest MAE, which means it is the most accurate on average.

b) Why the DNN baseline is important.: The poor performance of the plain DNN is not a weakness in the report; it is a useful scientific finding. It shows that a neural network without physics guidance is not enough for this problem and that the improvement from PIML is real, not incidental.

B. Relative Improvement

These percentages make the novelty visible. Compared to the Random Forest baseline, the Residual PIML reduces MAE by nearly 27%, and compared to the DNN baseline, it reduces MAE by more than 75%.

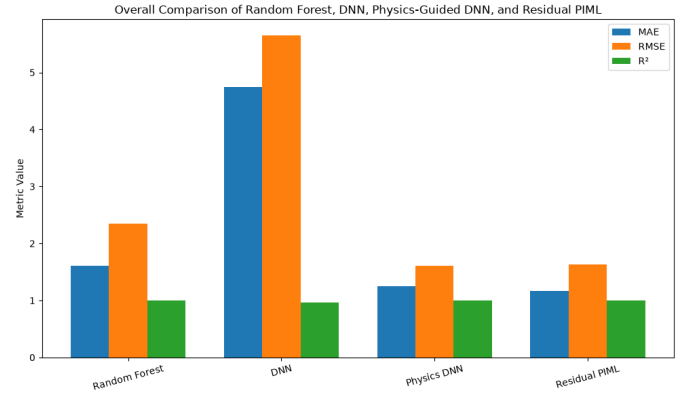


Fig. 4. Overall error comparison across the four model families: Random Forest, conventional neural network, physics-guided neural network, and residual physics-informed learning. The key trend is that adding physics guidance closes most of the performance gap between the weak plain neural baseline and the stronger tree-based baseline, while the residual formulation gives the lowest MAE by making the network learn only the correction to a physics estimate.

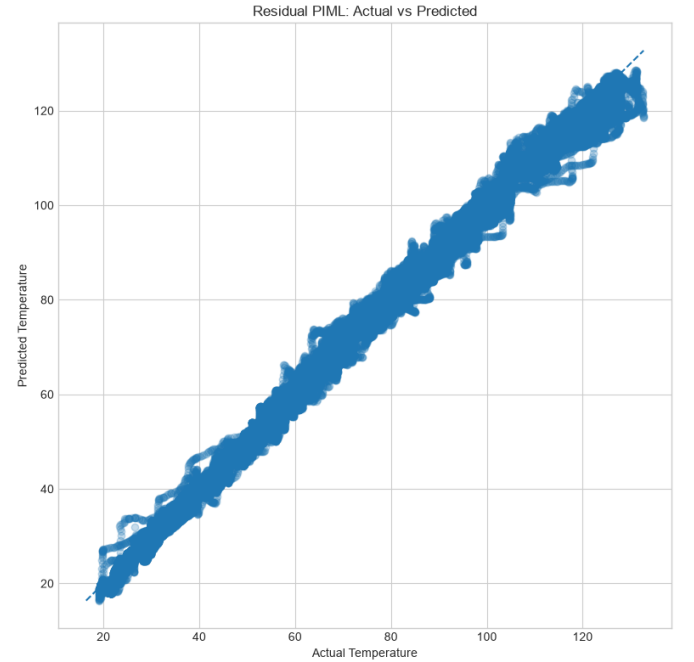


Fig. 5. Actual-versus-predicted stator winding temperature for the Residual PIML model. Points concentrated near the ideal diagonal indicate low bias across the operating range, while deviations from the diagonal identify regimes where the learned residual correction still underestimates or overestimates thermal response.

C. Why These Improvements Matter

The absolute error values are already small, but in thermal management even a one-degree reduction can matter because temperature thresholds are safety-critical. A more accurate predictor supports earlier warning, better protection, and more reliable deployment.

TABLE II
RELATIVE IMPROVEMENT OF PHYSICS-INFORMED MODELS

Model	MAE vs RF (%)	MAE vs DNN (%)	RMSE vs RF (%)	RMSE vs DNN (%)
Physics-Guided DNN	22.23	73.69	31.83	71.65
Residual PIML	26.90	75.27	30.49	71.09

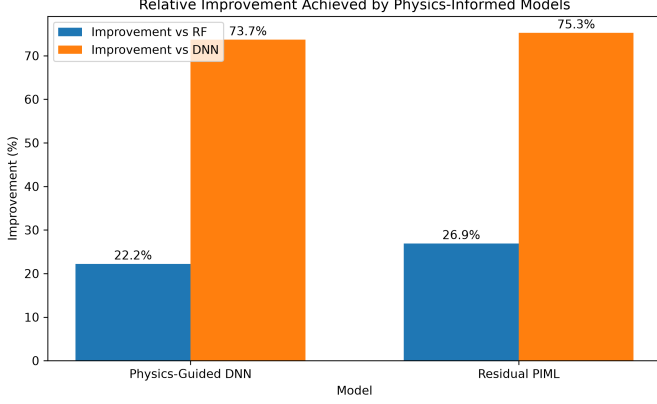


Fig. 6. Relative improvement of the physics-informed models over the Random Forest and DNN baselines. The large MAE reductions against the DNN baseline show that physics constraints prevent the neural model from learning unstable purely statistical mappings, while the gains over Random Forest demonstrate that the improvement is not only due to model capacity but also to physically meaningful problem formulation.

VII. PROFILE-WISE ANALYSIS

This section is essential because it proves the model is not only good on the global test set.

a) *How to interpret this table.*: Some profiles are easy for all models, while others are much harder. The plain DNN fails badly on several profiles, while the physics-guided models remain much more stable. This is exactly what the report should highlight: physics improves generalization when the operating regime changes.

b) *Important example.*: On profile 26, the DNN error is 7.4206°C , while the PG-DNN error drops to 0.9982°C . That is a dramatic improvement and a strong argument for physics-guided training.

VIII. STRESS TEST ANALYSIS

a) *Interpretation.*: High-stress conditions are where a naive predictor can become unsafe. Here the plain DNN again performs poorly, while the residual physics-informed model remains accurate. This supports the claim that the method is more suitable for real-world motor monitoring than a pure data-only model.

IX. UNCERTAINTY-AWARE ANALYSIS

Uncertainty analysis complements point-estimate accuracy by indicating when a thermal prediction should be trusted and when it should trigger a conservative safety margin. This is especially important for predictive maintenance because uncertain predictions near thermal limits should be handled

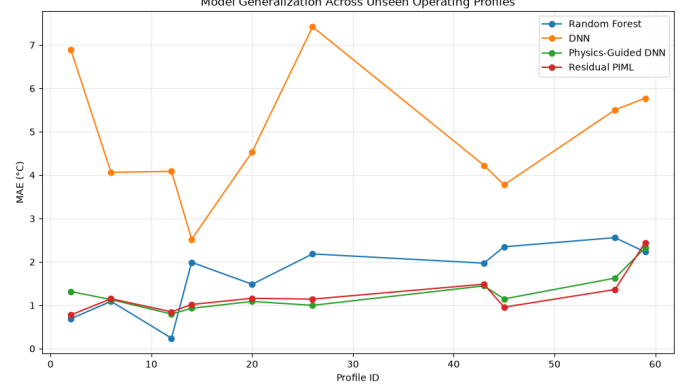


Fig. 7. Profile-wise MAE across selected operating profiles. This view tests generalization under regime shift: the plain DNN exhibits large profile-dependent error spikes, whereas the physics-guided models remain more stable, indicating that physics-informed features and losses transfer better to profiles not represented by a simple random split.

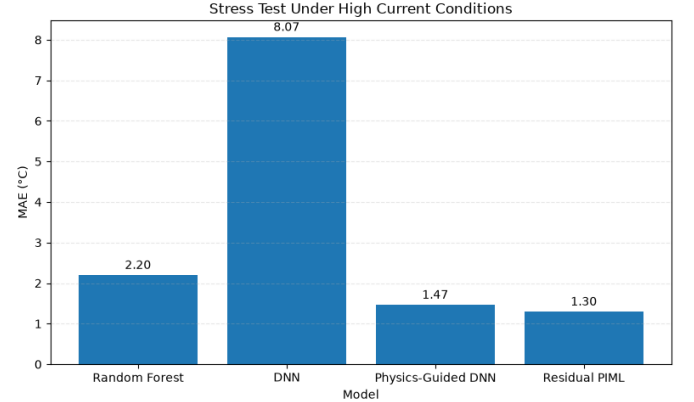


Fig. 8. High-current stress-test performance. This regime is thermally critical because copper loss scales approximately with current squared; the lower errors of the physics-informed models show that explicitly representing current-driven heating improves robustness where unsafe underprediction would be most damaging.

more cautiously than equally accurate predictions in familiar operating regimes.

X. EXPLAINABILITY AND WHY THE MODEL WORKS

A good research report should not stop at performance numbers. It should explain why the model behaves well. In this project, the strongest explanatory factors are current magnitude, coolant temperature, torque, and motor speed. This is physically sensible: current drives copper losses, coolant temperature changes the thermal boundary condition, and torque and speed influence load-dependent heating.

TABLE III
PROFILE-WISE MAE ON SELECTED OPERATING PROFILES

Profile ID	Samples	RF MAE	DNN MAE	PG-DNN MAE	Residual PIML MAE
2	19357	0.6877	6.9005	1.3169	0.7746
6	40388	1.0903	4.0650	1.1309	1.1495
12	21942	0.2405	4.0884	0.7966	0.8490
14	18598	1.9903	2.5216	0.9302	1.0180
20	43971	1.4842	4.5346	1.0881	1.1614
26	16666	2.1834	7.4206	0.9982	1.1413
43	8443	1.9714	4.2283	1.4463	1.4862
45	17142	2.3477	3.7770	1.1455	0.9560
56	33123	2.5586	5.5050	1.6262	1.3621
59	7475	2.2290	5.7754	2.3247	2.4459

TABLE IV
STRESS REGIME EVALUATION

Regime	Samples	RF MAE	DNN MAE	PG-DNN MAE	Residual PIML MAE
High Current	26720	2.2003	8.0699	1.4658	1.2983
High Mechanical Power	26720	1.9948	8.0823	1.3412	1.1386

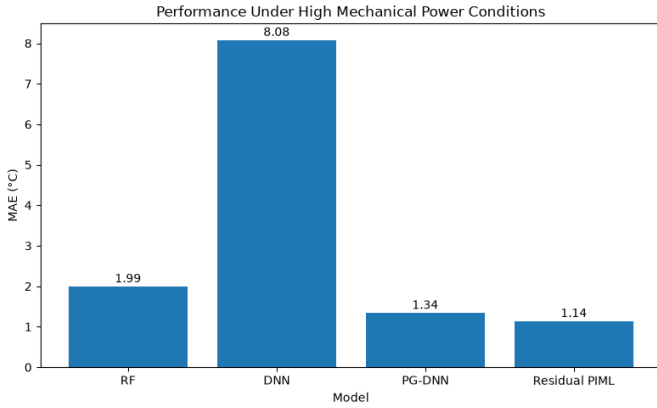


Fig. 9. High-mechanical-power stress-test performance. The figure evaluates whether the models remain accurate when torque and speed jointly indicate elevated load demand; the Residual PIML result is especially important because it suggests that the physics estimate plus learned correction remains reliable under load-intensive operating conditions.

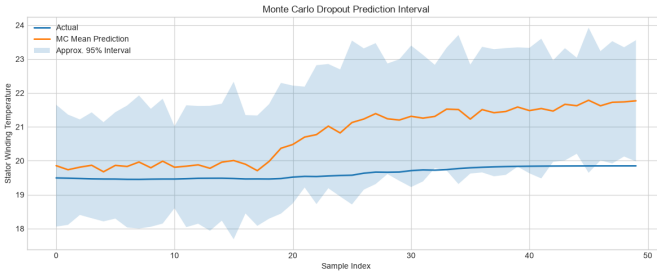


Fig. 10. Uncertainty-aware Residual PIML prediction. A narrow band around accurate predictions indicates confident operation in familiar regimes, while wider intervals flag regions where thermal estimates should be treated cautiously; this is useful for predictive maintenance because uncertainty can be converted into conservative warning thresholds.

If a feature importance method such as SHAP is too slow in a local environment, permutation importance can be used

instead. The point of the explainability section is to verify that the model is learning physically meaningful relationships rather than arbitrary correlations.

XI. NOVELTY

The novelty of this work is not simply that a neural network was trained on PMSM data. The novelty lies in the way the problem is formulated and tested.

- 1) **Physics-guided feature engineering:** current magnitude, voltage magnitude, electrical power, and mechanical power are introduced to reflect thermal behavior more directly.
- 2) **Physics-constrained training:** the loss function is augmented so that training respects physically reasonable temperature behavior.
- 3) **Residual PIML formulation:** instead of learning temperature from scratch, the model learns the residual error beyond a physics estimate.
- 4) **Robust evaluation:** the model is tested not only on a standard split but also under unseen profiles and stress regimes.
- 5) **Quantitative proof of benefit:** the proposed models are shown to outperform both Random Forest and conventional DNN baselines in MAE, RMSE, and R^2 .

XII. REAL-WORLD DEPLOYMENT PERSPECTIVE

The practical use case is predictive maintenance. In a deployment system, live sensor data from a motor would be fed to the trained model, which would output a temperature estimate. That estimate can then be converted into an operational health label such as *Healthy*, *Warning*, or *Critical*. This makes the work more than a benchmark exercise; it becomes a usable monitoring pipeline.

A. Why this is important

A model that predicts temperature accurately can trigger early intervention before thermal damage occurs. This can protect insulation, reduce downtime, and improve system reliability.

XIII. LIMITATIONS

The report should also be honest about what was not done.

- The physics model is simplified and not a full finite-element thermal simulation.
- Temporal sequence modeling was not fully exploited in the main pipeline.
- The residual physics formulation could be extended further using LSTM or transformer-based sequence models.
- Additional calibration against real deployment measurements would strengthen external validity.

XIV. FUTURE WORK

Possible extensions include:

- Physics-Informed LSTM for time-series thermal dynamics.
- Uncertainty-aware deployment on embedded or edge hardware.
- Digital twin integration for online thermal prediction.
- Remaining Useful Life estimation using the same thermal features.
- Fault diagnosis and anomaly detection using physics-informed embeddings.

XV. CONCLUSION

This report presented a physics-informed machine learning framework for PMSM stator winding temperature prediction. The main technical contribution is the Residual PIML model, which combines a physics-based thermal estimate with neural residual correction. The experimental results show clear and consistent improvement over a Random Forest baseline and a conventional DNN baseline. The final model achieved $MAE = 1.1719^{\circ}C$, $RMSE = 1.6336^{\circ}C$, and $R^2 = 0.9972$, while also demonstrating strong behavior under unseen profiles and high-stress operating conditions.

The main scientific conclusion is that physics is not just a nice addition; it materially improves prediction quality, generalization, and robustness. That makes the approach more credible for real-world predictive maintenance than a data-only model.

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